

RESEARCH INTERESTS

Soft matter; Polymer physics; Functional polymer hybrid materials; Biomolecular interfaces; Hierarchical structure; Computation-informed material design;

EDUCATION

Daegu Il Science High School, Korea High School Diploma	2018.02 – 2020.02
Pohang University of Science and Technology (POSTECH), Korea B.Sc., Chemistry GPA: 4.08/4.3 (U.S. Equivalent: 3.92/4.0) – ranked 1 st in class	2020.02 – 2025.02
TU Wien, Austria Exchange student	2023.10 – 2024.02
Nanyang Technological University, Singapore Summer exchange student	2024.06 – 2024.07

RESEARCH EXPERIENCE

Center for Self-assembly and Complexity (Prof. Kimoon Kim) POSTECH, Korea
Undergraduate Researcher 2022.06 – 2023.02; Resumed 2025.10 – present

- Presented a poster at the 130th General Meeting of the Korean Chemical Society (KCS).
- Led a project on programmable spatiotemporal control of chemical processes via multi-frequency acoustic stimulation for out-of-equilibrium polymer materials (manuscript in preparation).
- Investigated microgel-based polymer systems to develop pH-responsive interfacial patterning.

Smart Soft Materials Lab (Prof. Youn Soo Kim) POSTECH, Korea
Undergraduate Researcher 2024.09 – 2024.12

- Synthesized lignin-based hydrogels by engineering cation- π interactions and conducted mechanical (rheology, tensile, lap shear) and chemical characterization to optimize for bio-adhesive applications.
- Developed skills in polymer formulation, crosslinking control, and assessment of structure-property relationships in soft-matter materials.

Max Planck Institute for Polymer Research (Prof. Mischa Bonn) Mainz, Germany
Intern (Korea-Germany Junior Research Fellowship Program) 2023.08 – 2024.05

- Collaborated with international researchers and participated in scientific discussions at VeKNI, strengthening cross-disciplinary scientific communication skills.
- Elucidated the hydrogen bond asymmetry and anticorrelation in liquid water by integrating MD simulations (GROMACS) with 2D-IR spectroscopic data, successfully reproducing water's structural anomalies.
- Analyzed temperature-dependent IR spectroscopy and improved data processing workflows (MATLAB/Origin) for reproducible analysis.

Computational Energy/Bio Soft Material Lab (Prof. Chang Yun Son) POSTECH, Korea
Undergraduate Researcher 2023.02 – 2023.06

- Performed MD simulations (OpenMM) to investigate image-charge effects on ionic liquid systems (EMIM/BF₄ and BMIM/BF₄) confined in graphene electrodes, revealing enhanced charge separation at varying potentials.
- Gained proficiency in atomistic simulation techniques (WHAM/Umbrella Sampling for PMF calculations, protein folding simulations, and water-model validation).

ADDITIONAL RESEARCH TRAININGS

Quantum Dot Bioconjugation for Targeted Imaging POSTECH, Korea
Winter Research Program 2022.01 – 2022.02

- Synthesized antibody-conjugated core-shell quantum dots (CdSe/ZnS, AgInS/ZnS) and developed surface functionalization strategies for targeted bio-imaging.

Functional Polymer Thin Films & Bio-interfaces

KAIST, Korea

Technical Training

2025.08 – 2025.10

- Trained in iCVD fabrication of polymer nanocoatings, utilizing FTIR and water contact angle for characterization, and evaluating bio-interfacial properties via *in vitro* assays (cell culture, PCR, and immunofluorescence staining).

TEACHING

Student Mentoring Program | POSTECH

2021.03 – 2023.06

- Taught General Chemistry and Organic Chemistry to freshmen and sophomores, and mentored academic planning.

LEADERSHIP

The POSTECH Times | POSTECH

2020.04 – 2023.03

Editor-in-chief (2022.08 – 2023.03); Head of the Reporting Office (2022.02 – 2022.07); Reporter

- Oversaw monthly publication as Editor-in-Chief; managed editorial planning and supervised reporting teams.
- Led an exclusive interview with a Nobel laureate and edited scientific feature articles for the university community.
- Strengthened organizational leadership through team management and cross-department coordination.

Management Strategy Student Association (MSSA) | POSTECH

2021.03 – 2021.12

- Collaborated on case analyses of technology/entertainment companies, enhancing analytical and strategic thinking.
- Mentored new members to improve presentation and communication skills.

Chemistry Student Council | POSTECH

2022.01 – 2022.12

Treasurer

- Managed a \$6,000 departmental budget and organized welfare and academic events for 90 students.
- Facilitated a career exploration event by coordinating faculty–student discussions.

Student Council (Entire Student Body) | POSTECH

2020.03 – 2021.07

Financial department

- Managed budgeting and accounting for the university-wide Student Council through cross-departmental collaboration.
- Organized community-building events and supported student engagement during the pandemic.

HONORS & AWARDS

Jigok Scholarship (Full Tuition Scholarship) | POSTECH

2020.02 – 2024.02

Chemistry Scholarship of Eminent Members | Chemistry department, POSTECH

2021.09

Lee In-sook Scholarship (Top Academic Excellence Scholarship) | Chemistry department, POSTECH

2022.07

Research Support Fund | Max Planck POSTECH/Korea Research Initiative

2023.08 – 2024.01

MIRAE ASSET Global Exchange Scholarship | Mirae Asset Park Hyeon Joo Foundation

2023.07

Jung Sung-gi Award (Graduation with Highest Honor) | Chemistry department, POSTECH

2025.02

President's Award | Korean Society for Biochemistry and Molecular Biology

2025.02

SKILLS

Languages

English (proficient); Korean (native); German (elementary)

Programming

Python; C; MATLAB; Linux (basic); VMD

Techniques

MD simulation (GROMACS, OpenMM; WHAM, Umbrella Sampling)

Spectroscopy (IR, UV-Vis, PL, NMR)

Materials (rheometry, mechanical testing, hydrogel synthesis, iCVD; bioconjugation, staining)

REFERENCES

Mischa Bonn | Director, Max Planck Institute for Polymer Research, Mainz, Germany

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Chang Yun Son | Assistant Professor, Department of Chemistry, Seoul National University, Seoul, Republic of Korea

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